A Biofidelic Composite Femur Surrogate for Orthopaedic Training, and the Population the Standard Model Excludes

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Abstract. Orthopaedic surgical training depends on surrogates that reproduce the tactile response of bone, and a surrogate that is mechanically wrong teaches the wrong force. This report identifies a composite material system that replicates the human femur under compressive loading for a 50th-percentile adult male, then extends the same analysis to a population the standard model excludes: postmenopausal women with osteoporosis. Cortical and trabecular properties were derived by linear regression of the elastic region of provided uniaxial compression and torsion data ($R^2 = 1.000$), giving compressive moduli of 17.40 and 1.24 GPa respectively. A CT-derived femur model with a 5.0 mm cortical shell around a trabecular core was loaded at 2,619 N, three times body weight, at 7° of adduction to represent single-leg stance. The fourteenfold stiffness mismatch places roughly 93 % of the load in the cortical shell; all midshaft stresses remain sub-yield and peak von Mises stress occurs at the femoral neck, consistent with clinical fracture patterns. Glass-fibre reinforced epoxy and 20 PCF rigid polyurethane foam match healthy cortical and trabecular bone. Re-running the analysis with osteoporotic postmenopausal geometry and properties drives minimum safety factors toward 1.0, which is what fragility fracture risk looks like inside a simulation, and implies a second surrogate specification rather than a scaled version of the first.

Keywords. surgical simulation; bone surrogate; finite element analysis; composite materials; osteoporosis; design bias; femoral neck

1 Introduction

Orthopaedic surgical training requires realistic tactile feedback. Suture pads and bone surrogates let trainees rehearse technique before operating on a patient, and the value of the rehearsal depends entirely on whether the surrogate loads and fails the way bone does. A surrogate that is too stiff teaches a trainee to under-apply force; one that is too compliant teaches the opposite.

This project has two objectives. The first is conventional: identify a composite material system that mechanically replicates the human femur under compressive loading for a 50th-percentile adult male. The second is the reason the project is worth writing up. The 50th-percentile male is the default design target across orthopaedic hardware and training tools, and it excludes a large share of the population that actually presents for femoral surgery. The analysis is therefore carried through a second time for postmenopausal women with osteoporosis, and the two specifications are reported side by side.

2 Derived mechanical properties

Uniaxial compression and torsion stress–strain data were provided for cortical and trabecular bone from a 28-year-old male donor. Properties were obtained by linear regression of the elastic region, which fitted essentially exactly ($R^2 = 1.000$). Elastic modulus E and shear modulus G are the regression slopes. Poisson's ratio was estimated from the isotropic relation $\nu = E/(2G) - 1$, giving values consistent with the literature [1,2]. Yield stress was taken at a 0.2 % offset for cortical bone and at the linear-to-nonlinear transition for trabecular bone, whose stress–strain response has no clean offset construction.

Table 1. Mechanical properties derived from the provided compression and torsion data. Values marked ≈ are construction-dependent.

Property	Cortical	Trabecular	Unit
Compressive modulus, E	17.40	1.24	GPa
Shear modulus, G	6.69	0.53	GPa
Poisson's ratio, ν (estimated)	0.30	0.17	–
Compressive yield stress, σ_y	≈90	≈8.5	MPa
Ultimate compressive stress, σ_u	139.3	11.93	MPa
Shear yield stress, τ_y	≈63.6	≈1.8	MPa

3 Geometry and loading

The femur was modelled from a CT-derived mesh as a cortical shell around a trabecular core, scaled to a length of 480 mm to match reported average adult male femoral length [3,4]. Cortical thickness was set to 5.0 mm, giving an inner trabecular diameter of approximately 18 mm [5]. The resulting cortical-to-total cross-sectional area ratio is about 47 %, against a reported value near 51.5 % [6], and the midshaft outer dimensions of approximately 38.7×35.0 mm give a cross-sectional area of about 1,064 mm^2.

A 50th-percentile adult male has a mass of approximately 89 kg, or 873 N of body weight [7]. Hip contact forces reach roughly three times body weight during single-leg stance [8], so a compressive load of 2,619 N was applied to the femoral head at a 7° adduction angle to reproduce physiological alignment [4].

4 Simulation results

A static finite element simulation was run in Autodesk Fusion. The cortical shell ($E = 17{,}400$ MPa, $\nu = 0.30$) and trabecular core ($E = 1{,}240$ MPa, $\nu = 0.17$) were modelled as bonded bodies, with a fixed constraint applied distally and the compressive load applied proximally.

Table 2. Static finite element results under 2,619 N single-leg stance loading.

Result	Value	Note
Mean cortical axial stress, midshaft	4.85 MPa	vs. $\sigma_y \approx 90$ MPa
Mean trabecular axial stress, midshaft	0.35 MPa	vs. $\sigma_y \approx 8.5$ MPa
Peak von Mises stress, femoral neck	14–16 MPa	consistent with Papini *et al.* [4]
Minimum safety factor, cortical neck	5.7	against yield
Minimum safety factor, trabecular	>12	against yield
Total axial deformation	0.134 mm	$\delta = \sigma L/E$, $L = 480$ mm
Cortical load share	93 %	stiffness-weighted parallel model

Because the modulus ratio $E_{\text{cortical}}/E_{\text{trabecular}}$ is approximately 14, the cortical shell carries roughly 93 % of the applied load [4]. All stresses remain well below yield, confirming that physiological single-leg stance is not a failure condition in healthy bone. Peak stress occurs at the femoral neck rather than the midshaft, which matches the clinical fracture distribution and is the first indication that the neck, not the shaft, is the region a surrogate has to get right.

5 Bias in the 50th-percentile male standard

Designing to a 50th-percentile male introduces a specific bias: the standard reflects young male populations and excludes a large fraction of orthopaedic patients [9]. Postmenopausal women with osteoporosis are the clearest omission. Women account for approximately 80 % of osteoporosis cases, and hip fractures fall disproportionately on women over 65 [10]. The relevant differences are structural as well as material:

- **Shorter femora**, roughly 402–424 mm against 480 mm in males [11].
- **Reduced bone density**, with 1–2 % annual loss following menopause [10].
- **Lower structural strength** from cortical thinning and increased porosity [12].
- **Trabecular degradation**, with substantially reduced moduli [13].
- **Thinner cortex**, approximately 2–3 mm against 4.5–5.0 mm [5].

A training tool built to healthy male bone therefore feels overly stiff. A surgeon who calibrates force and judges fixation behaviour on that tool is calibrating against the wrong material, in the direction that matters most for the patients most likely to fracture.

6 Proposed surrogate materials

Cortical analogue. Glass-fibre reinforced epoxy ($E = 15$–20 GPa, strength 130–160 MPa, $\nu \approx 0.30$) is already an industry standard cortical bone analogue and brackets the derived cortical modulus of 17.40 GPa [4].

Trabecular analogue. 20 PCF rigid polyurethane foam ($E = 1.0$–1.5 GPa, strength 8–14 MPa) replicates healthy trabecular bone [13].

Table 3. Two surrogate specifications. The osteoporotic column is a proposed parameter set for simulation, not a validated build.

Parameter	Healthy 50th-percentile male	Osteoporotic postmenopausal female
Length	480 mm	395 mm [11]
Diameter	28 mm	25 mm [5]
Cortical thickness	5.0 mm	2.5 mm [5]
Cortical E	17.4 GPa	15–17 GPa [12]
Trabecular E	1.24 GPa	0.2–0.5 GPa [13]
Applied load	2,619 N	1,700 N [8]
Cortical analogue	Glass-fibre epoxy	*to reconcile* †
Trabecular analogue	20 PCF PU foam	3–5 PCF PU foam [13]

† The working notes list a 10 PCF polyurethane foam as the osteoporotic cortical analogue, which is inconsistent with the 15–17 GPa cortical modulus in the row above by roughly two orders of magnitude. Osteoporotic cortical bone loses thickness and gains porosity far faster than it loses tissue-level modulus, so the correct substitution is almost certainly the same glass-fibre epoxy at reduced wall thickness rather than a foam. This row is left open pending a decision.

Under the osteoporotic parameter set, the combination of a thinner cortex, weaker trabecular core and reduced but not proportionally reduced load produces higher stresses and minimum safety factors approaching 1.0. That is the numerical signature of fragility fracture risk, and it is the result that justifies building two surrogates rather than one.

7 Limitations

- **Isotropic assumption.** Bone is anisotropic; treating it as isotropic will overestimate transverse stiffness.
- **Single-specimen data.** Properties derive from one 28-year-old male donor and capture none of the biological variability that matters for a population-level design target.
- **Static loading only.** Fatigue and cyclic effects are excluded, which is a significant omission for osteoporotic bone specifically.
- **Bonded interface.** A perfectly bonded shell-core interface will overestimate load transfer between the two materials.
- **Microstructural mismatch.** Polyurethane foams reproduce bulk modulus and strength but not trabecular architecture, so tactile realism during drilling and screw purchase is not guaranteed by matching bulk properties.
- **Unresolved units.** One source range for degraded trabecular modulus appears in the working notes in MPa where GPa is implied; the figure in Table 3 follows the GPa reading.

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Document note. Two citations were corrected while preparing this document: reference [3] is Aitken (2021) in *Osteology*, not Sabharwal in *Diagnostics*, and reference [6] appeared in *HOMO – Journal of Comparative Human Biology*, not the *Journal of Forensic and Legal Medicine*. Two internal inconsistencies in the source notes are flagged rather than silently resolved: the osteoporotic cortical analogue row in Table 3 and the unit ambiguity noted under Limitations. Reference [11] could not be independently verified to volume and page level and should be checked before circulation.